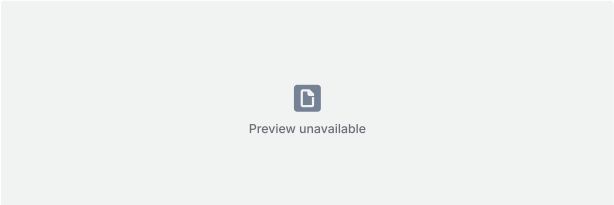
Assessment 2 - SQL

26. Write a query in SQL to find the sum, average, and number of order quantity for those orders whose ids are 43659 and 43664 and product id starting with '71'. Return SalesOrderID, OrderNumber,ProductID, OrderQty, sum, average, and number of order quantity.

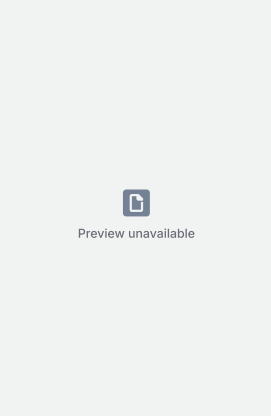
```sql
SELECT SalesOrderID, ProductID, OrderQty,
    SUM(OrderQty) OVER() AS TotalQuantity,
    AVG(OrderQty) OVER() AS AvgQuantity,
    COUNT(OrderQty) OVER()  AS NoOfOrders
FROM Sales.SalesOrderDetail
WHERE SalesOrderID IN (43659, 43664) AND ProductID LIKE '71%'
ORDER BY ProductID ASC;
```



---

27. Write a query in SQL to retrieve the total cost of each salesorderID that exceeds 100000. Return SalesOrderID, total cost

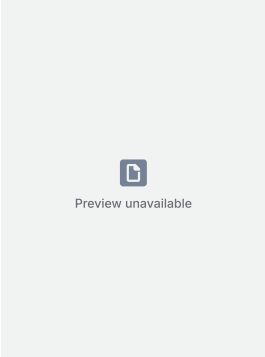
```sql
SELECT SalesOrderID, TotalDue
FROM Sales.SalesOrderHeader
WHERE TotalDue >100000;
```



---

28. Write a query in SQL to retrieve products whose names start with 'Lock Washer'. Return product ID, and name and order the result set in ascending order on product ID column.

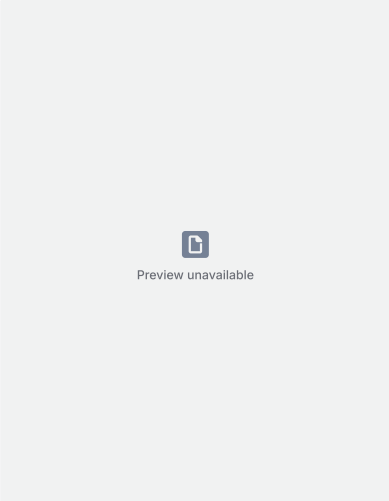
```sql
SELECT ProductID, Name
FROM Production.Product
WHERE Name LIKE 'Lock Washer%'
ORDER BY ProductID ASC;
```



---

29. Write a query in SQL to fetch rows from product table and order the result set on an unspecified column listprice. Return product ID, name, and color of the product.

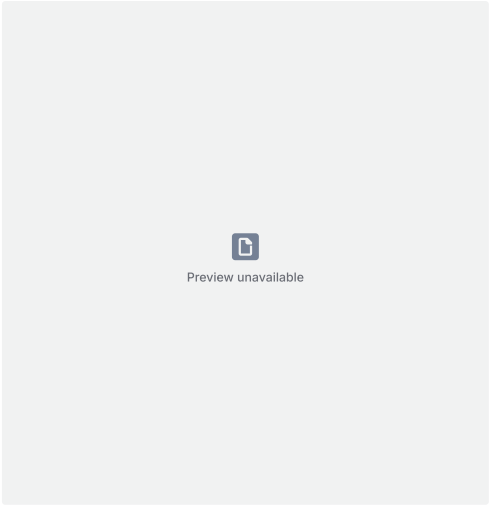
```sql
SELECT ProductID, Name, ISNULL(Color,'') as Color
FROM Production.Product
ORDER BY ListPrice;
```



---

30. Write a query in SQL to retrieve records of employees. Order the output on year (default ascending order) of hiredate. Return BusinessEntityID, JobTitle, and HireDate.

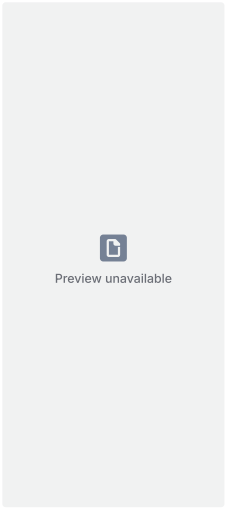
```sql
SELECT BusinessEntityID, JobTitle, HireDate
FROM HumanResources.Employee
ORDER BY HireDate ASC;
```



---

31. Write a query in SQL to retrieve those persons whose last name begins with letter 'R'. Return lastname, and firstname and display the result in ascending order on firstname and descending order on lastname columns

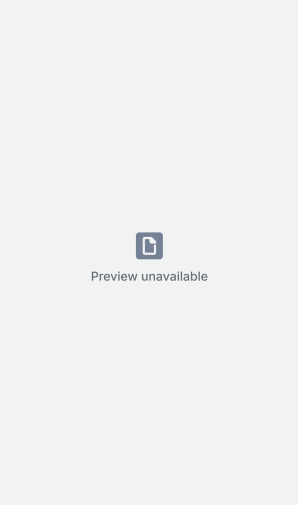
```sql
SELECT LastName, FirstName
FROM Person.Person
WHERE LastName LIKE 'R%'
ORDER BY FirstName ASC, LastName DESC;
```



---

32. Write a query in SQL to ordered the BusinessEntityID column descendingly when SalariedFlag set to 'true' and BusinessEntityID in ascending order when SalariedFlag set to 'false'. Return BusinessEntityID, SalariedFlag columns

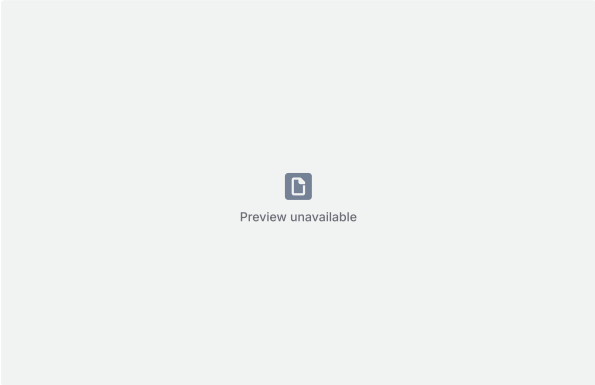
```sql
SELECT BusinessEntityID,
    IIF(SalariedFlag = 1, 'True' , 'False') AS SalariedFlag
FROM HumanResources.Employee
ORDER BY
    IIF(SalariedFlag = 1, -BusinessEntityID, BusinessEntityID);
```



---

33. Write a query in SQL to set the result in order by the column TerritoryName when the column CountryRegionName is equal to 'United States' and by CountryRegionName for all other rows

```
SELECT SP.BusinessEntityID, P.LastName, ST.Name AS TerritoryName,
CR.Name AS CountryRegionName
FROM Sales.SalesPerson AS SP
    JOIN Person.Person AS P ON SP.BusinessEntityID = P.BusinessEntityID
    JOIN Sales.SalesTerritory AS ST ON SP.TerritoryID = ST.TerritoryID
    JOIN Person.CountryRegion AS CR ON ST.CountryRegionCode =
CR.CountryRegionCode
ORDER BY
    IIF(CR.Name = 'United States', ST.Name, CR.Name);
```

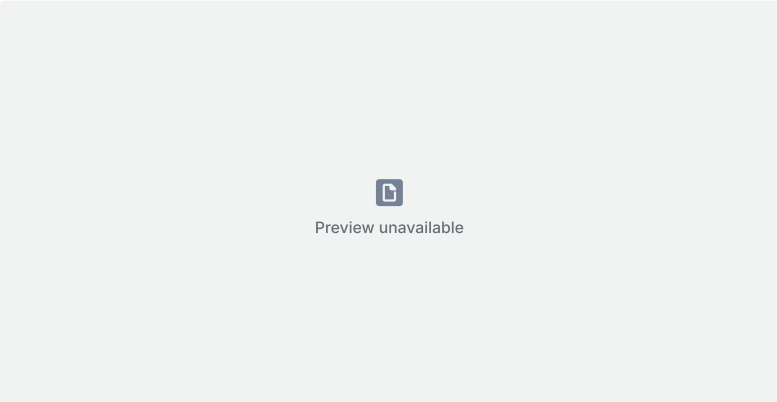


---

34. Write a query in SQL to find those persons who lives in a territory and the value of salesytd except 0. Return first name, last name,row number as 'Row Number', 'Rank', 'Dense Rank' and NTILE as 'Quartile', salesytd and postalcode. Order the output on postalcode column.

```
SELECT P.FirstName, P.LastName,
    ROW_NUMBER() OVER(ORDER BY A.PostalCode) AS RowNumber,
    RANK() OVER(ORDER BY A.PostalCode) AS Rank,
    DENSE_RANK() OVER(ORDER BY A.PostalCode) AS DenseRank,
    NTILE(4) OVER(ORDER BY SP.SalesYTD) AS Quartile,
    SP.SalesYTD, A.PostalCode
FROM Sales.SalesPerson AS SP
    JOIN Sales.SalesTerritory AS ST ON SP.TerritoryID = ST.TerritoryID
    JOIN Person.Person AS P ON SP.BusinessEntityID =
P.BusinessEntityID
    JOIN Person.BusinessEntityAddress AS BEA ON P.BusinessEntityID =
BEA.BusinessEntityID
    JOIN Person.Address AS A ON BEA.AddressID = A.AddressID
```

```
WHERE SP.SalesYTD > 0
ORDER BY A.PostalCode ASC
```

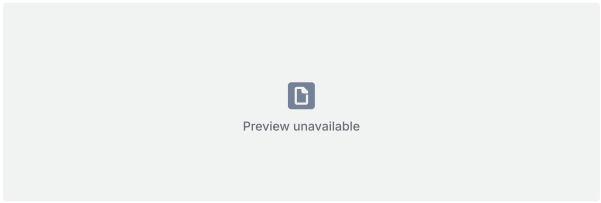


---

35. From the following table write a query in SQL to skip the first 10 rows from the sorted result set and return all remaining rows

```
/*SELECT DepartmentID, Name, GroupName FROM
    (SELECT DepartmentID, Name, GroupName, ROW_NUMBER() OVER(ORDER BY DepartmentID) AS RowNumber
    FROM HumanResources.Department) AS D
WHERE RowNumber > 10
ORDER BY DepartmentID*/

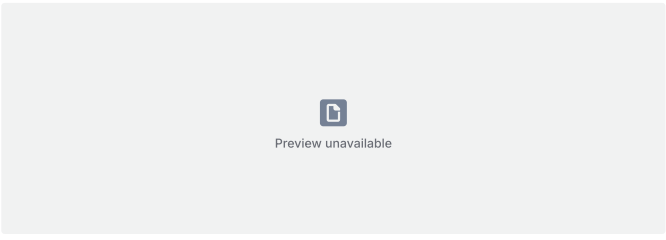
SELECT DepartmentID, Name, GroupName
FROM HumanResources.Department
ORDER BY DepartmentID OFFSET 10 ROWS
```



---

36. Write a query in SQL to skip the first 5 rows and return the next 5 rows from the sorted result set

```
SELECT DepartmentID, Name, GroupName
FROM HumanResources.Department
ORDER BY DepartmentID
    OFFSET 5 ROWS
    FETCH NEXT 5 ROWS ONLY
```

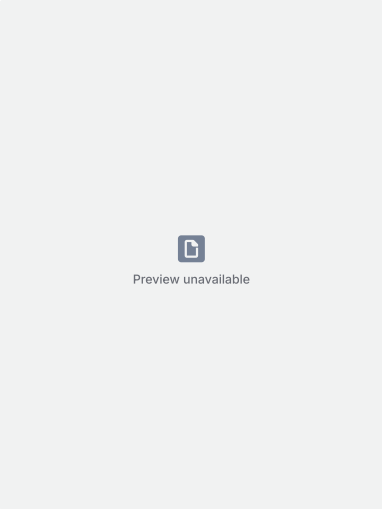


---

37. From the following table write a query in SQL to list all the products that are Red or Blue in color. Return name, color and listprice.Sorts this result by the column listprice

```
SELECT Name, Color, ListPrice
```

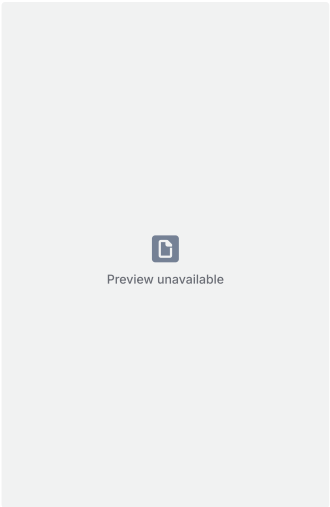
```
FROM Production.Product
WHERE Color IN ('Blue', 'Red')
ORDER BY ListPrice;
```



---

38. Create a SQL query from the SalesOrderDetail table to retrieve the product name and any associated sales orders. Additionally, it returns any sales orders that don't have any items mentioned in the Product table as well as any products that have sales orders other than those that are listed there. Return product name, salesorderid. Sort the result set on product name column

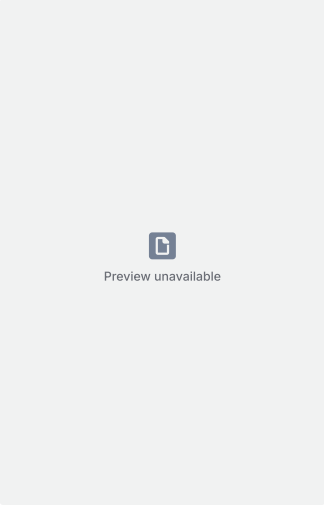
```
SELECT P.Name, S.SalesOrderID
FROM Sales.SalesOrderDetail S
    FULL OUTER JOIN Production.Product P ON S.ProductID = P.ProductID
ORDER BY P.Name;
```



---

39. Write a SQL query to retrieve the product name and salesorderid. Both ordered and unordered products are included in the result set

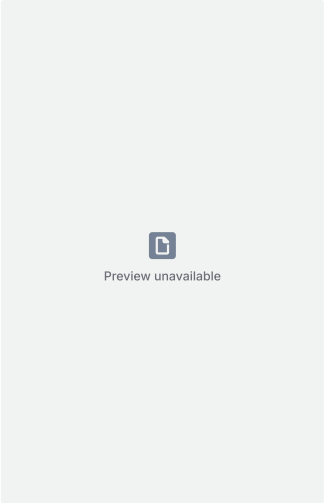
```
SELECT P.Name, S.SalesOrderID
FROM Sales.SalesOrderDetail S
    RIGHT JOIN Production.Product P ON P.ProductID = S.ProductID
ORDER BY P.Name;
```



---

40. Write a SQL query to get all product names and sales order IDs. Order the result set on product name column

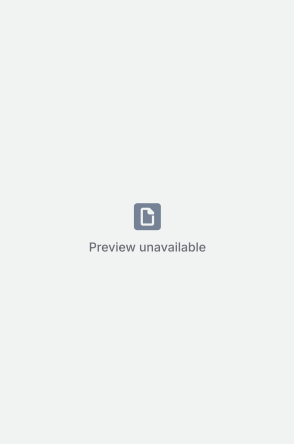
```sql
SELECT P.Name, S.SalesOrderID
FROM Sales.SalesOrderDetail S
    INNER JOIN Production.Product P ON S.ProductID = P.ProductID
ORDER BY P.Name;
```



---

41. Write a SQL query to retrieve the territory name and BusinessEntityID. The result set includes all salespeople, regardless of whether or not they are assigned a territory

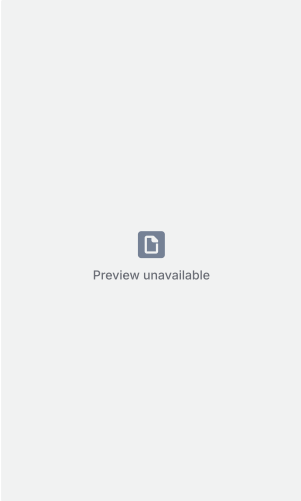
```sql
SELECT ISNULL(T.Name,'') AS Territory, P.BusinessEntityID
FROM Sales.SalesPerson P
    LEFT JOIN Sales.SalesTerritory T ON P.TerritoryID = T.TerritoryID
ORDER BY P.BusinessEntityID;
```



---

42. Write a query in SQL to find the employee's full name (firstname and lastname) and city from the following tables. Order the result set on lastname then by firstname

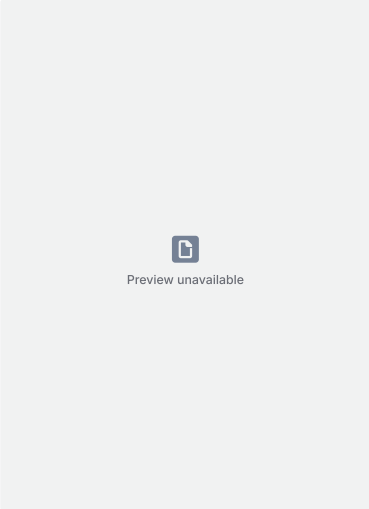
```
SELECT (P.FirstName+' '+P.LastName) AS Name, A.City
FROM Person.Person P
    JOIN Person.BusinessEntityAddress BEA ON BEA.BusinessEntityID =
P.BusinessEntityID
    JOIN Person.Address A ON A.AddressID = BEA.AddressID
ORDER BY P.LastName, P.FirstName;
```



---

43. Write a SQL query to return the businessentityid,firstname and lastname columns of all persons in the person table (derived table) with persontype is 'IN' and the last name is 'Adams'. Sort the result set in ascending order on firstname. A SELECT statement after the FROM clause is a derived table.

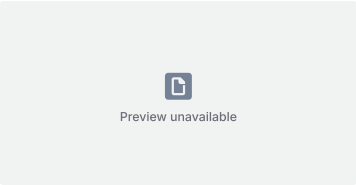
```
SELECT BusinessEntityID, FirstName, LastName
FROM (
        SELECT BusinessEntityID, FirstName, LastName
        FROM Person.Person
        WHERE PersonType = 'IN' AND LastName = 'Adams'
    ) AS PT
ORDER BY PT.FirstName ASC;
```



---

44. Create a SQL query to retrieve individuals from the following table with a businessentityid inside 1500, a lastname starting with 'Al', and a firstname starting with 'M'.

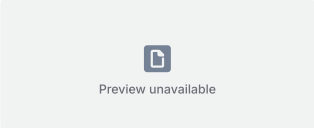
```sql
SELECT BusinessEntityID, FirstName, LastName
FROM Person.Person
WHERE BusinessEntityID < 1500
    AND LastName LIKE 'Al%'
    AND FirstName LIKE 'M%'
ORDER BY BusinessEntityID;
```



---

45. Write a SQL query to find the productid, name, and colour of the items 'Blade', 'Crown Race' and 'AWC Logo Cap' using a derived table with multiple values

```sql
SELECT ProductID, Name, Color
FROM (
        SELECT *
        FROM Production.Product) AS DT
WHERE Name IN ('Blade','Crown Race','AWC Logo Cap')
```

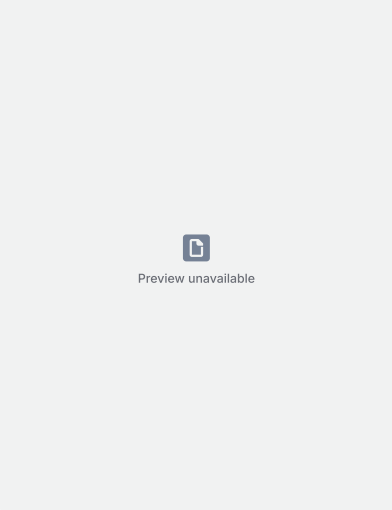


---

46. Create a SQL query to display the total number of sales orders each sales representative receives annually. Sort the result set by SalesPersonID and then by the date component of the orderdate in ascending order. Return the year component of the OrderDate, SalesPersonID, and SalesOrderID

```sql
SELECT SalesPersonID,
    COUNT(SalesPersonID) AS TotalSales,
    YEAR(OrderDate) AS SalesYear
FROM Sales.SalesOrderHeader
WHERE SalesPersonID IS NOT NULL
GROUP BY SalesPersonID, YEAR(OrderDate)
```

```
ORDER BY SalesPersonID, YEAR(OrderDate)
```

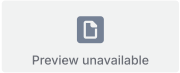


---

47. From the following table write a query in SQL to find the average number of sales orders for all the years of the sales representatives.

```
WITH SalesPerPerson AS
(
    SELECT SalesPersonID,
        COUNT(SalesPersonID) AS TotalSales
    FROM Sales.SalesOrderHeader
    WHERE SalesPersonID IS NOT NULL
    GROUP BY SalesPersonID
)

SELECT AVG(TotalSales) AS AverageSales FROM SalesPerPerson
```



---

48. Write a SQL query on the following table to retrieve records with the characters green_ in the LargePhotoFileName field. The following table's columns must all be returned

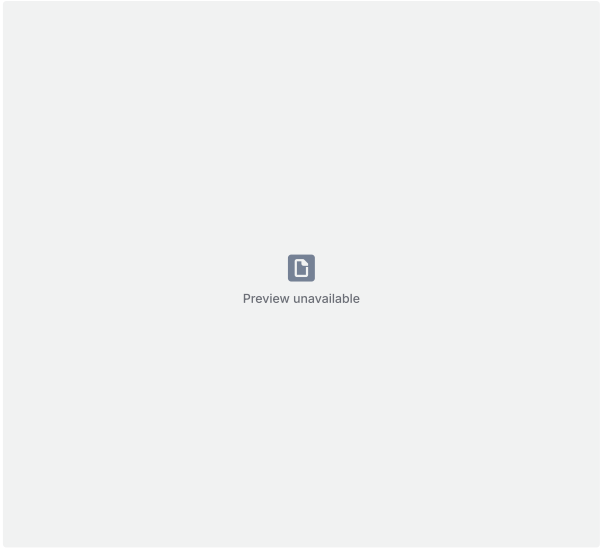
```
SELECT *
FROM Production.ProductPhoto
WHERE LargePhotoFileName LIKE '%green_%'
```



---

49. Write a SQL query to retrieve the mailing address for any company that is outside the United States (US) and in a city whose name starts with Pa. Return Addressline1, Addressline2, city, postalcode, countryregioncode columns.

```
SELECT A.AddressLine1, A.AddressLine2, A.City, A.PostalCode,
SP.CountryRegionCode
FROM Person.Address A
    JOIN Person.StateProvince SP ON A.StateProvinceID =
SP.StateProvinceID
WHERE SP.Name <> 'United States' AND A.City LIKE 'Pa%
```



---

50. Write a query in SQL to fetch first twenty rows. Return jobtitle, hiredate. Order the result set on hiredate column in descending order

```sql
SELECT TOP 20 JobTitle, HireDate
FROM HumanResources.Employee
ORDER BY HireDate DESC;
```

